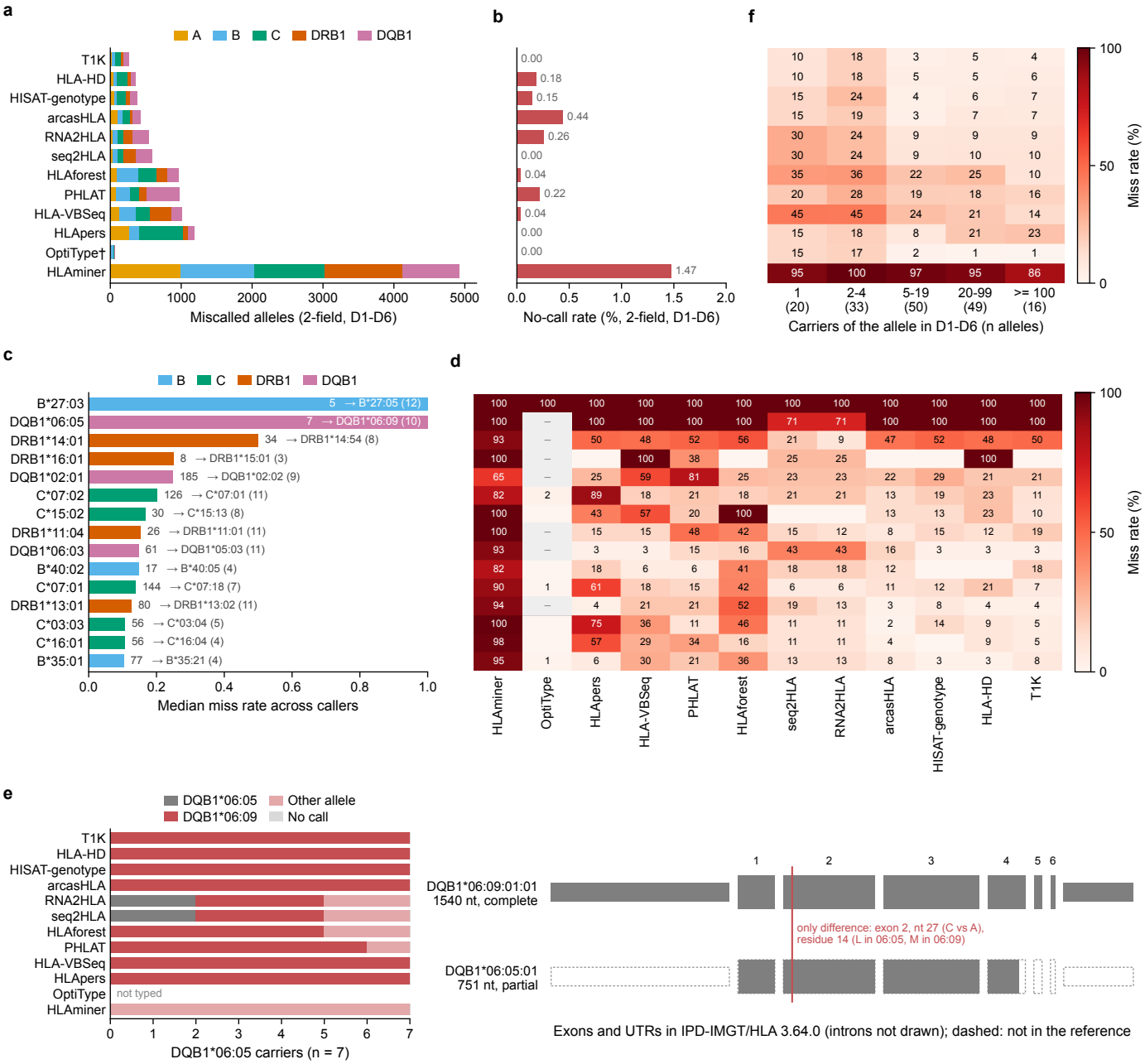

a
A
B
C
DRB1
DQB1
T1K
HLA-HD
HISAT-genotype
arcasHLA
RNA2HLA
seq2HLA
HLAforest
PHLAT
HLA-VBSeq
HLApers
OptiType†
HLAminer
0
1000
2000
3000
4000
5000
Miscalled alleles (2-field, D1-D6)
b
0.00
0.18
0.15
0.44
0.26
0.00
0.04
0.22
0.04
0.00
0.00
1.47
0.0
0.5
1.0
1.5
2.0
No-call rate (%, 2-field, D1-D6)
f
10 18 3 5 4
10 18 5 5 6
15 24 4 6 7
15 19 3 7 7
30 24 9 9 9
30 24 9 10 10
35 36 22 25 10
20 28 19 18 16
45 45 24 21 14
15 18 8 21 23
15 17 2 1 1
95 100 97 95 86
1 (20)
2-4 (33)
5-19 (50)
20-99 (49)
>= 100 (16)
Carriers of the allele in D1-D6 (n alleles)
100
50
0
Miss rate (%)
c
B
C
DRB1
DQB1
B*27:03 5 → B*27:05 (12)
DQB1*06:05 7 → DQB1*06:09 (10)
DRB1*14:01 34 → DRB1*14:54 (8)
DRB1*16:01 8 → DRB1*15:01 (3)
DQB1*02:01 185 → DQB1*02:02 (9)
C*07:02 126 → C*07:01 (11)
C*15:02 30 → C*15:13 (8)
DRB1*11:04 26 → DRB1*11:01 (11)
DQB1*06:03 61 → DQB1*05:03 (11)
B*40:02 17 → B*40:05 (4)
C*07:01 144 → C*07:18 (7)
DRB1*13:01 80 → DRB1*13:02 (11)
C*03:03 56 → C*03:04 (5)
C*16:01 56 → C*16:04 (4)
B*35:01 77 → B*35:21 (4)
0.0
0.2
0.4
0.6
0.8
1.0
Median miss rate across callers
d
100 100 100 100 100 100 100 100 100 100 100 100
100 – 100 100 100 100 71 71 100 100 100 100
93 – 50 48 52 56 21 9 47 52 48 50
100 – 100 38 25 25 100
65 – 25 59 81 25 23 23 22 29 21 21
82 2 89 18 21 18 21 21 13 19 23 11
100 43 57 20 100 13 13 23 10
100 – 15 15 48 42 15 12 8 15 12 19
93 – 3 3 15 16 43 43 16 3 3 3
82 18 6 6 41 18 18 12 18
90 1 61 18 15 42 6 6 11 12 21 7
94 – 4 21 21 52 19 13 3 8 4 4
100 75 36 11 46 11 11 2 14 9 5
98 57 29 34 16 11 11 4 9 5
95 1 6 30 21 36 13 13 8 3 3 8
HLAminer
OptiType
HLApers
HLA-VBSeq
PHLAT
HLAforest
seq2HLA
RNA2HLA
arcasHLA
HISAT-genotype
HLA-HD
T1K
100
50
0
Miss rate (%)
e
DQB1*06:05
DQB1*06:09
Other allele
No call
T1K
HLA-HD
HISAT-genotype
arcasHLA
RNA2HLA
seq2HLA
HLAforest
PHLAT
HLA-VBSeq
HLApers
OptiType
not typed
HLAminer
0
1
2
3
4
5
6
7
DQB1*06:05 carriers (n = 7)
1
2
3
4
5
6
DQB1*06:09:01:01
1540 nt, complete
only difference: exon 2, nt 27 (C vs A),
residue 14 (L in 06:05, M in 06:09)
DQB1*06:05:01
751 nt, partial
Exons and UTRs in IPD-IMGT/HLA 3.64.0 (introns not drawn); dashed: not in the reference